

NOVA KAKHOVKA, Ukraine

WMO: 338690

Lat: 46.7892N

Lon: 33.3586E

Elev: 82

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	10.7	0.9	5.8	8.2	4.8	7.1	12.2	24.0	27.2	21.3	22.3	8.2	20	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	89.7	69.7	87.0	68.5	84.7	67.7	72.2	84.3	70.8	83.2	69.3	81.3	5.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.3	104.1	77.2	66.5	97.7	76.0	64.9	92.5	74.7	35.9	84.7	34.6	83.3	33.4	81.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	17.1	14.7	1.8	94.2	5.5	3.3	-2.1	96.6	-5.3	98.5	-8.4	100.3	-12.4	102.7		
			1.0	75.2	5.4	2.8	-2.8	77.2	-6.0	78.9	-9.0	80.5	-12.9	82.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	29.6	31.5	39.2	51.1	61.1	68.6	73.5	73.1	63.6	52.0	39.7	31.9	
	DBStd	17.59	9.70	8.91	8.66	6.46	6.30	5.37	4.90	5.65	7.09	8.65	8.63	8.06	
	HDD50	2526	632	518	341	64	4	0	0	0	2	80	322	563	
	HDD65	5741	1097	938	801	416	153	25	3	6	107	408	760	1027	
	CDD50	3022	0	1	6	99	347	557	728	716	411	144	12	1	
	CDD65	760	0	0	0	1	31	132	266	257	67	6	0	0	
	CDH74	6343	0	0	0	8	201	942	2332	2398	446	16	0	0	
	CDH80	1957	0	0	0	1	17	225	749	858	106	1	0	0	
Wind	WSAvg	6.6	7.1	7.4	7.7	6.8	5.7	5.3	5.4	5.6	5.7	7.4	7.3	7.7	
Precipitation	PrecAvg	16.5	1.2	1.1	1.1	1.2	1.6	2.1	1.5	1.3	1.5	1.2	1.3	1.3	
	PrecMax	24.8	2.8	3.1	2.3	3.1	4.5	4.0	3.7	3.9	4.6	3.6	3.1	3.6	
	PrecMin	10.1	0.2	0.2	0.2	0.1	0.2	0.8	0.1	0.1	0.2	0.1	0.0	0.2	
	PrecStd	3.6	0.7	0.6	0.6	0.9	1.0	0.8	1.1	0.9	1.1	0.7	0.8	0.8	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.7	56.8	63.9	74.6	82.1	88.8	92.7	93.0	88.3	76.1	61.9	53.9
MCWB			45.0	47.9	52.2	58.5	64.6	68.8	71.8	70.2	69.5	61.3	56.3	50.3	
2%		DB	45.8	49.6	58.1	69.7	79.2	85.4	89.3	90.0	82.8	71.8	57.5	48.5	
		MCWB	43.4	45.5	48.7	56.2	62.9	67.8	69.8	69.4	66.4	60.1	54.2	45.0	
5%		DB	43.3	45.6	54.0	66.3	76.2	82.5	86.7	87.6	79.3	67.5	54.8	45.1	
		MCWB	41.2	42.2	45.8	53.7	61.3	66.6	69.0	68.2	64.4	58.4	52.0	42.7	
10%		DB	40.7	42.2	50.2	63.0	73.2	79.6	84.3	85.0	75.3	63.9	51.3	42.2	
		MCWB	38.8	39.6	43.5	52.4	60.2	65.9	68.0	67.6	63.1	57.0	48.6	40.6	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	46.5	49.2	52.8	60.2	66.8	71.6	74.7	73.0	71.0	65.0	57.1	51.3
			MCDB	47.8	54.0	62.0	72.5	77.4	83.7	85.3	86.6	82.6	70.4	59.9	51.9
	2%	WB	43.9	45.8	49.6	57.2	65.2	69.7	73.0	71.5	68.1	62.4	55.0	45.4	
		MCDB	45.4	49.5	56.6	67.1	75.0	80.6	83.8	85.9	79.3	68.3	57.1	47.9	
	5%	WB	41.4	42.6	47.0	55.3	63.6	68.4	71.5	70.1	66.2	60.3	52.1	43.0	
		MCDB	42.9	45.0	52.4	64.7	72.9	79.3	83.1	84.2	74.7	65.8	54.6	45.0	
	10%	WB	38.9	39.8	44.9	53.3	61.9	67.1	70.0	68.7	64.5	57.9	49.0	40.5	
		MCDB	40.5	42.0	49.0	61.0	71.0	77.2	81.4	81.9	73.1	63.1	50.9	42.1	
	Mean Daily Temperature Range	5% DB	MDBR	9.0	10.4	12.6	16.3	17.4	16.7	18.4	18.8	16.9	13.9	9.4	8.6
			MCDBR	11.2	15.3	19.8	22.2	21.0	20.9	21.2	21.5	21.1	18.1	12.9	11.5
MCWBR			8.9	10.9	11.2	10.7	8.8	8.2	8.1	8.8	8.7	9.4	9.7	8.7	
5% WB		MCDBR	10.2	14.2	18.6	20.6	18.6	18.1	19.1	19.7	18.4	16.4	12.1	10.9	
		MCWBR	8.6	10.8	11.4	10.9	8.5	8.1	8.6	9.2	9.4	10.0	9.8	8.4	
Clear-Sky Solar Irradiance	taub	0.339	0.368	0.395	0.444	0.419	0.421	0.445	0.442	0.413	0.376	0.357	0.340		
	taud	2.291	2.229	2.203	2.094	2.255	2.279	2.222	2.217	2.263	2.346	2.331	2.313		
	Ebn at Noon	231	248	259	257	269	269	260	255	251	243	222	215		
	Edh at Noon	25	33	39	47	42	41	43	42	36	29	24	22		
All-Sky Solar Radiation	RadAvg	330	608	1000	1525	1923	2032	2036	1831	1313	810	387	268		
	RadStd	56	79	104	130	161	137	121	95	135	97	57	45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.31	N/A	N/A	+1.85	N/A	N/A	-198	-275	+223	+148

Nomenclature: See separate page